

CANADA POP

Canada's Linguistic Future

A Demographic Simulation, 1971–2150

This report presents the methodology, data sources, parameters, and findings of an agent-based model projecting Canada's population by language group across 13 provinces and territories from 1971 to 2150.

Classification

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Contents

Executive Summary	2
1 Methodology	2
1.1 Model Design	2
1.2 Why the Decline Slows: The Plateau Mechanism	3
1.3 Calibration and Validation	4
1.4 Scenarios	4
2 Data Sources	4
3 Model Parameters	5
3.1 Starting Population — 1971 Census	5
3.2 Immigration Volume	6
3.3 Linguistic Composition of Immigrants	6
3.4 Language Transfer Rates	7
3.5 Interprovincial Migration and Language Multipliers	8
4 Key Findings	8
4.1 Finding 1 — National Francophone Share Declines but Plateaus	8
4.2 Finding 2 — The Dilution Gap Narrows Toward Equilibrium	9
4.3 Finding 3 — Francophones Grow in Absolute Numbers	10
4.4 Finding 4 — Quebec Holds; the Rest of Canada Does Not	11
4.5 Finding 5 — New Brunswick: Slow but Steady Erosion	13
4.6 Finding 6 — Immigration Dilution Drives the Decline; Language Transfer Is Secondary . .	13
4.7 Finding 7 — Stabilising the Share Would Require Doubling the Immigrant Francophone Fraction	14
5 Conclusions & Strategic Implications	15

Executive Summary

At a Glance

- Canada's national francophone share declines from **19%** in 2025 to between **8% and 12%** by 2150 across all scenarios, but the rate of decline continuously slows.
- The decline is a **denominator effect**: the francophone population grows in absolute terms from 7.5 million to 12–13 million over the same period.
- **Immigration composition, not volume, is the primary lever.** Increasing intake without raising the francophone share of immigrants accelerates dilution.
- Francophones outside Quebec face a **far steeper relative decline**, compounded by high rates of language transfer to English.
- Stabilising the national francophone share would require roughly **19% of all immigrants to be francophone** — nearly double current federal targets and far above actual achievement.

This report documents Canada Pop, a discrete-time agent-based simulation of Canada's demographic evolution by language group. The model integrates four forces — international immigration, interprovincial migration, natural increase, and language transfer — calibrated to Statistics Canada census data from 1971 to 2025 and projected to 2150 under three scenarios. All scenarios agree on the direction of travel; they differ substantially on how fast the francophone share falls and where it ultimately stabilises.

The most consequential finding is structural: because the francophone share of immigrants to Canada ($\approx 6\text{--}14\%$) is far below the current population share ($\approx 19\%$), each year's intake pulls the national share downward. This gap is the engine of decline, and closing it entirely is not achievable under any realistic near-term policy trajectory. What is achievable is moderating the speed of convergence, and the High scenario demonstrates meaningfully better outcomes from a sustained, ambitious francophone immigration strategy.

1. Methodology

1.1. Model Design

Canada Pop is a **discrete-time, agent-based model**. Each simulated agent represents 100 real people. Every agent has exactly two attributes: the province they live in (one of 13) and their language group (Francophone, Anglophone, or Allophone). The model starts from the 1971 Census and advances one year at a time through to 2150.

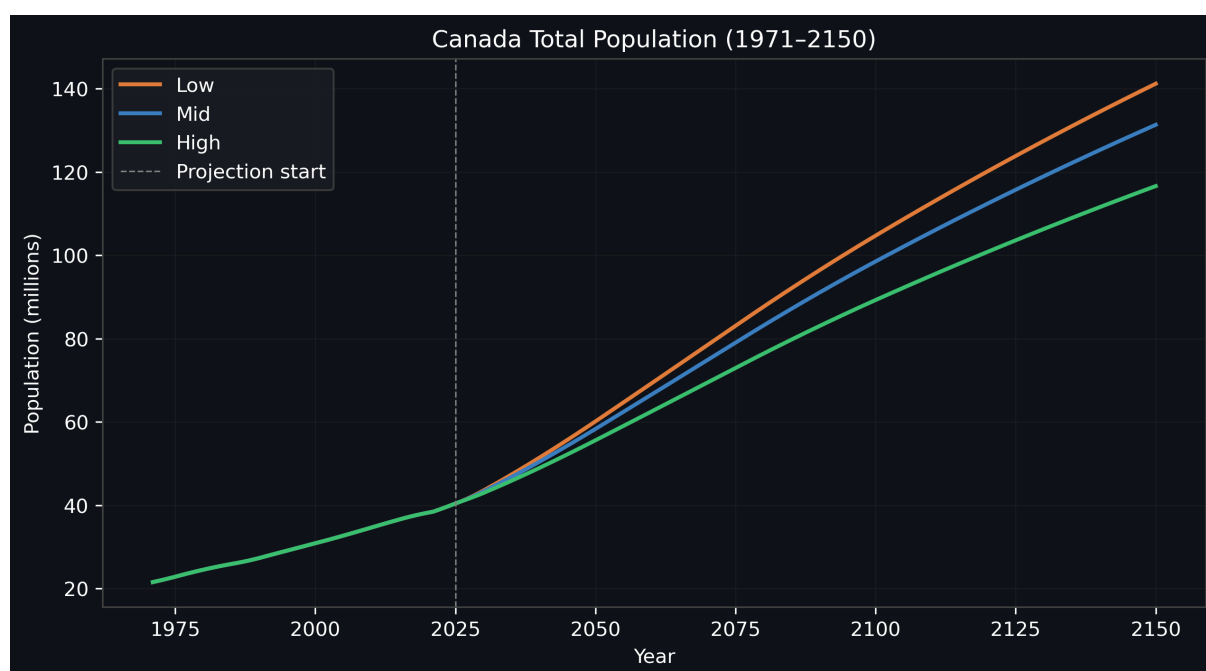


Figure 1. Simulated total population of Canada by scenario (1971–2150). The three scenarios are identical during the 1971–2025 calibration period and diverge post-2025 based on different immigration volumes. The Mid scenario is calibrated to reach approximately 100 million by 2100.

At each annual time step the model applies four processes in sequence:

1. **Language transfer** — a small fraction of agents switch language groups each year, using province-specific rates calibrated to historical census data. In Quebec, allophones shift toward French (reflecting Bill 101); outside Quebec, francophones shift toward English.
2. **Interprovincial migration** — agents move between provinces using historical flow rates, with language-group multipliers reflecting empirical asymmetries (e.g. Quebec francophones are highly immobile; Quebec anglophones historically left at $2.5\times$ the provincial average).
3. **Natural increase** — agents are added or removed to reflect net births minus deaths. New agents inherit the language distribution of their province. Most provinces are projected to have negative natural increase by the 2030s.
4. **International immigration** — new agents arrive each year, allocated to provinces and assigned a language based on the assumed composition of the immigrant intake.

1.2. Why the Decline Slows: The Plateau Mechanism

A key analytical result that shapes how all the projection curves look is the *self-decelerating* nature of immigration-driven share dilution.

Each year, immigration pulls the francophone share toward the francophone fraction of the immigrant intake. But the size of that annual pull is proportional to the gap between the current share and the immigrant share. As the national francophone share falls and narrows toward the immigrant fraction, the

pull weakens — the system automatically brakes. This produces the characteristic curved, plateauing shape of the projections rather than a straight-line decline.

Intuition. If immigrants are 14% francophone and the population is currently 18% francophone, immigration delivers a 4-point downward pull each year. But once the share has fallen to 15%, the pull is only 1 point. The closer the share gets to the immigrant fraction, the slower it moves. The decline decelerates continuously and the share approaches — but never crosses — the immigrant fraction as a floor.

1.3. Calibration and Validation

The period 1971–2025 uses empirical data and is identical across all three scenarios. The model is validated at census years (1971, 1991, 2001, 2011, 2021) against Statistics Canada figures for total population, national francophone share, and Quebec francophone share. The mid-scenario is calibrated to reproduce Canada's population reaching approximately 100 million by 2100, consistent with long-run IRCC planning assumptions.

1.4. Scenarios

Three scenarios govern the post-2025 projection period. They differ on two dimensions: the total volume of immigration and the francophone fraction of that intake.

Table 1. Scenario parameters applied from 2026 onward.

Parameter	Low	Mid	High
Immigration volume multiplier	× 1.10	× 1.00	× 0.85
Francophone share offset	−4.0 pp	0.0 pp	+4.0 pp
ROC francophone share (fixed)	1.0%	4.0%	8.9%
<i>Interpretation</i>	Pessimistic	Trend	Optimistic

The **Low** scenario represents a pessimistic outcome for French: higher immigration volumes combined with a reduced francophone intake share. The **High** scenario represents sustained federal commitment to francophone immigration targets with moderately reduced overall volumes. The **Mid** scenario is a trend continuation, holding existing patterns broadly constant.

2. Data Sources

All model inputs are stored as CSV files in the `data/` directory. Table 2 summarises the primary data sources used for each component.

Table 2. Primary data sources by model component.

Component	Source
Initial population (1971)	Statistics Canada, <i>1971 Census of Canada</i> . Population by province; language shares from “language spoken most often at home.”
Historical population benchmarks	Statistics Canada Census of Canada, 1971–2021 (quinquennial). StatCan quarterly estimate Q1 2025 ($\approx$ 41.5 million).
Francophone share targets	Statistics Canada Census 1971–2021 — French spoken most often at home, national and Quebec-specific series.
Natural increase rates	Statistics Canada CANSIM Table 17-10-0008: annual births, deaths, and net natural increase by province, 1971–2024.
Interprovincial migration flows	Statistics Canada CANSIM Table 17-10-0015: annual interprovincial migrants by origin and destination province.
Language transfer rates	Statistics Canada <i>Language Projections, 2011–2036</i> (Cat. 89-657-X) and OQLF data sheets.
Immigration volume (1971–2025)	IRCC annual permanent resident landings data and the 2025 immigration levels plan.
Immigrant francophone share — Quebec	Office québécois de la langue française (OQLF) and Ministère de l’Immigration, de la Francisation et de l’Intégration (MIFI), 2001–2025.
Immigrant francophone share — ROC	IRCC Annual Reports 2003–2025 (federal francophone immigration target data and achievement rates).

Post-2025 values in all input files are projections based on trend extrapolation of the above sources. They do not represent official Statistics Canada or Government of Canada forecasts.

3. Model Parameters

3.1. Starting Population — 1971 Census

The simulation is initialised from the 1971 Census. Table 3 shows the starting population and language composition for each province. Quebec is the only province with a francophone majority. New Brunswick at 31% is the only other province with a substantial francophone share.

Table 3. Initial population and language shares by province, 1971 Census.

Province	Population	Fr share	En share	Allo share
NL	522,000	0.3%	99.2%	0.5%
PE	112,000	4.0%	95.0%	1.0%
NS	789,000	2.5%	95.5%	2.0%
NB	635,000	31.0%	68.0%	1.0%
QC	6,028,000	80.8%	13.2%	6.0%
ON	7,703,000	4.6%	75.4%	12.0%
MB	988,000	4.5%	85.5%	11.0%
SK	926,000	1.8%	91.2%	7.0%
AB	1,628,000	1.4%	90.6%	8.0%
BC	2,185,000	0.8%	89.2%	10.0%
YT, NT, NU	53,000	mixed	—	—
Canada	21,569,000	25.7%	66.5%	7.8%

3.2. Immigration Volume

Annual immigrant landings follow historical IRCC data from 1971 to 2025. After 2025 the volume is projected under each scenario. The base trajectory (Mid scenario multiplier = 1.0) is shown in Table 4 at selected years.

Table 4. Annual immigration volume (base trajectory, mid scenario) at selected years.

Year	Landings	Year	Landings
1971	122,000	2025	500,000
1985	84,000	2030	700,000
2000	227,000	2050	910,000
2020	184,000	2100	960,000
2023	471,000	2150	900,000

The 2020 dip reflects COVID-19 border restrictions. The post-2030 ramp reflects continuation of the trajectory set by IRCC's 2025 levels plan, scaled to meet the 100-million-by-2100 calibration target in the Mid scenario.

3.3. Linguistic Composition of Immigrants

Quebec and the Rest of Canada operate under different selection regimes, producing very different linguistic profiles in the immigrant intake.

Table 5. Francophone share of immigrants by destination region at selected years (mid scenario). Post-2025 values are trend projections; scenario offsets and overrides are applied on top.

Year	Quebec fr share	ROC fr share
1971	35.0%	2.0%
2001	51.0%	3.0%
2016	60.5% (peak)	3.0%
2021	54.5%	4.5%
2025	56.0%	8.9%
2050	49.0%	8.2%
2100	41.5%	7.5%
2150	39.0%	7.0%

The Quebec series peaked in 2016 (60.5% per OQLF data), driven by Bill 101 enforcement and active Quebec selection of French-speaking immigrants. The subsequent decline reflects structural shifts in source-country composition. The ROC series rose sharply from 2022–2025 on a surge of francophone Sub-Saharan African arrivals, then stabilises below the federal government’s aspirational 12% target.

3.4. Language Transfer Rates

Every year a small fraction of agents shift language groups, reflecting intergenerational assimilation. Table 6 shows the most policy-relevant rates calibrated to the 2021 period.

Table 6. Annual language transfer rates at the 2021 calibration point. These are per-person annual probabilities of switching language group.

Region	From	To	Annual rate	Context
Quebec	Allo	French	1.6% / yr	Bill 101 French integration
Quebec	Allo	English	1.0% / yr	Declining English pull
Quebec	English	French	0.25% / yr	Modest Anglo-to-French shift
Quebec	French	English	0.05% / yr	Negligible
NB	French	English	0.6% / yr	Gradual Acadian erosion
ROC	French	English	1.8% / yr	≈41% cumulative lifetime
ROC	Allo	English	2.0% / yr	≈50% cumulative lifetime
ROC	Allo	French	0.06% / yr	Minimal francophone retention

The ROC francophone-to-English rate of 1.8% per year is the highest of any group and reflects the assimilation pressure faced by small francophone communities embedded in overwhelmingly anglophone environments.

3.5. Interprovincial Migration and Language Multipliers

Base migration flows are calibrated from StatCan data for three anchor periods (1971, 2000, 2024). Table 7 shows the language-specific multipliers that adjust the base out-migration rate for particular province-language combinations.

Table 7. Language-conditional out-migration multipliers (1.0 = provincial average).

Province	Language	Multiplier	Reason
Quebec	Francophone	0.30	Francophone Quebecers are highly immobile
Quebec	Anglophone	2.50	Post-1976 Anglo exodus
Quebec	Allophone	1.50	Above-average departure
Ontario	Francophone	1.50	Franco-Ontarians above-average mobile
NB	Francophone	0.70	Acadian community stickiness
Manitoba	Francophone	1.30	Small community, mobile
Alberta	Francophone	2.00	Oil-sector high mobility
BC	Francophone	1.80	Small dispersed community

4. Key Findings

4.1. Finding 1 — National Francophone Share Declines but Plateaus

Under all three scenarios, Canada's national francophone share falls continuously from approximately 19% in 2025 (Figure 2). By 2150 the share reaches 8% under the Low scenario, 9.5% under Mid, and 12% under High. The High scenario visibly flattens after 2100, approaching its equilibrium. Under Low, the decline continues more steeply throughout, reflecting the compounding of higher volume and lower francophone intake share.

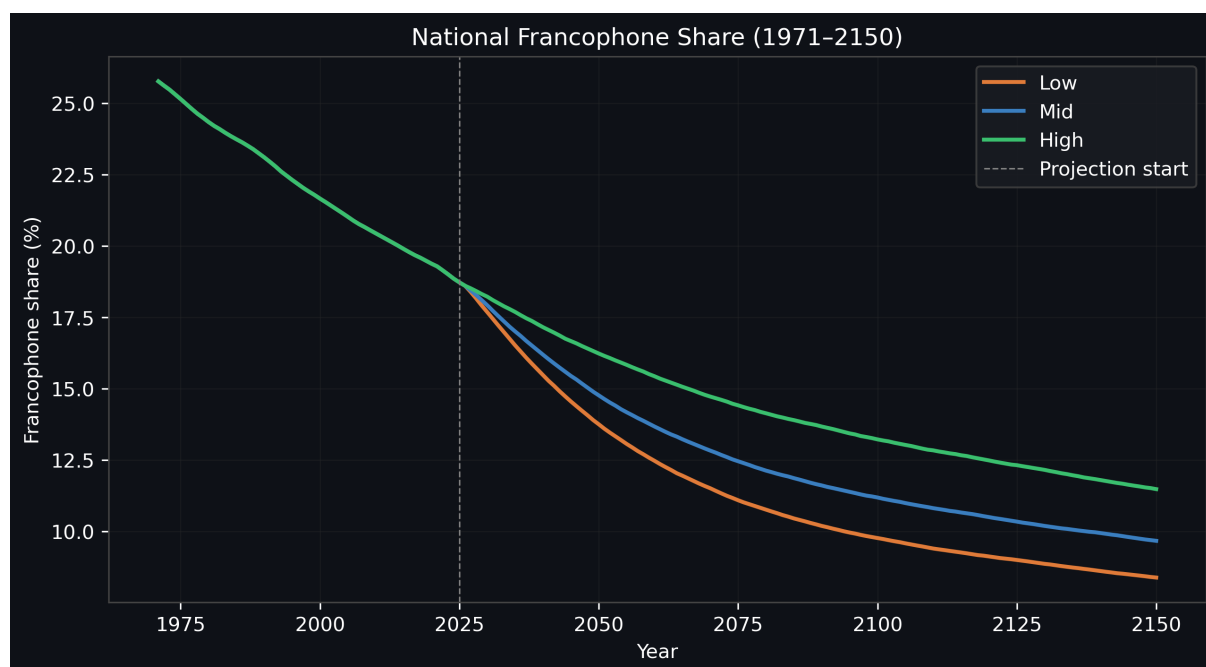


Figure 2. National francophone share of Canada's total population (1971–2150) under three scenarios. The dashed vertical line marks the end of the calibration period. All three scenarios reproduce the historical 1971–2025 decline and diverge thereafter.

The curves are visibly curved rather than straight. This is the plateau mechanism described in Section 1.2: each year's dilution force is proportional to the gap between the current share and the immigrant share, so as the share falls, the force weakens.

Key takeaway. The national francophone share will roughly halve from its 2025 level by 2150 in the Mid scenario. The difference between Low and High by 2150 is approximately 4 percentage points — a meaningful but bounded range of outcomes.

4.2. Finding 2 — The Dilution Gap Narrows Toward Equilibrium

Figure 3 overlays the national francophone share $f(t)$ (solid lines) with the effective francophone fraction of immigrants f_{imm} (dashed lines) for each scenario. The gap between solid and dashed is the engine driving the decline each year.

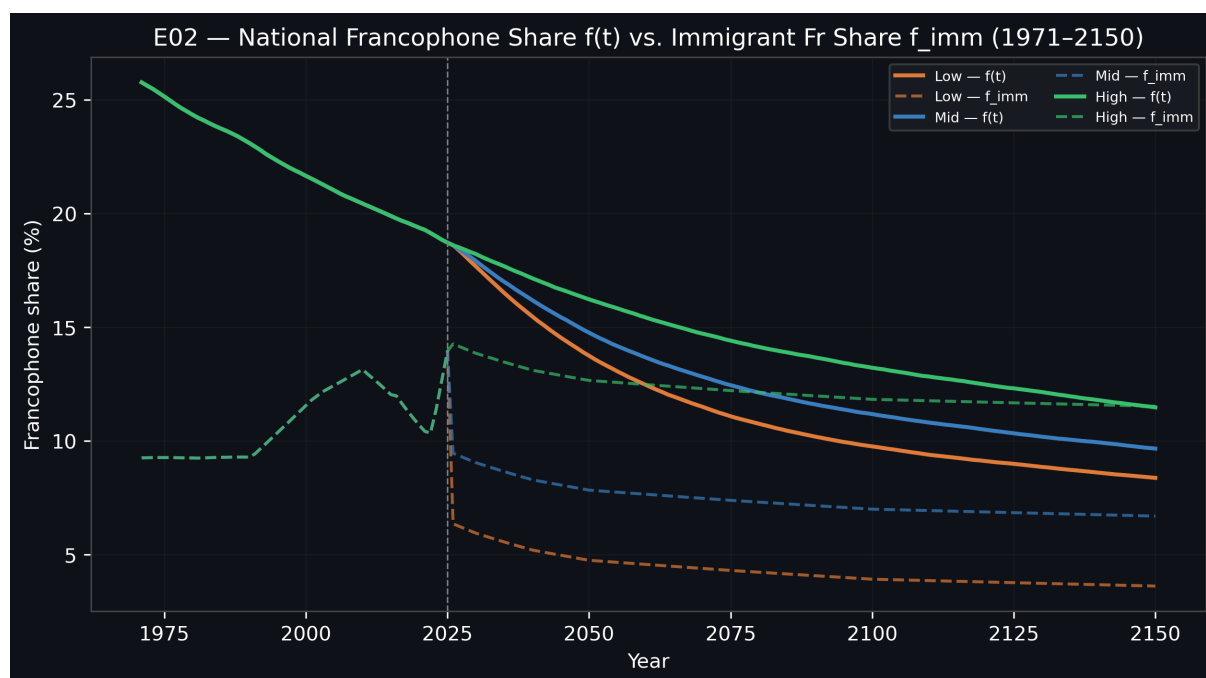


Figure 3. National francophone share $f(t)$ (solid) vs. effective francophone share of annual immigrants f_{imm} (dashed) by scenario. The solid lines chase the dashed lines from above; the gap between them drives the rate of decline.

The sharp drop in the dashed lines at 2025 reflects the scenario offsets taking effect. In the Low scenario, the immigrant francophone share is effectively just 6% after 2025 — far below the population share — explaining the steep initial decline. In the High scenario, f_{imm} is held near 12–14%, and the solid line approaches it slowly, producing the visible plateau after 2100.

4.3. Finding 3 — Francophones Grow in Absolute Numbers

Figure 4 shows that despite declining in share, the francophone population grows in absolute terms throughout the projection period under all scenarios.

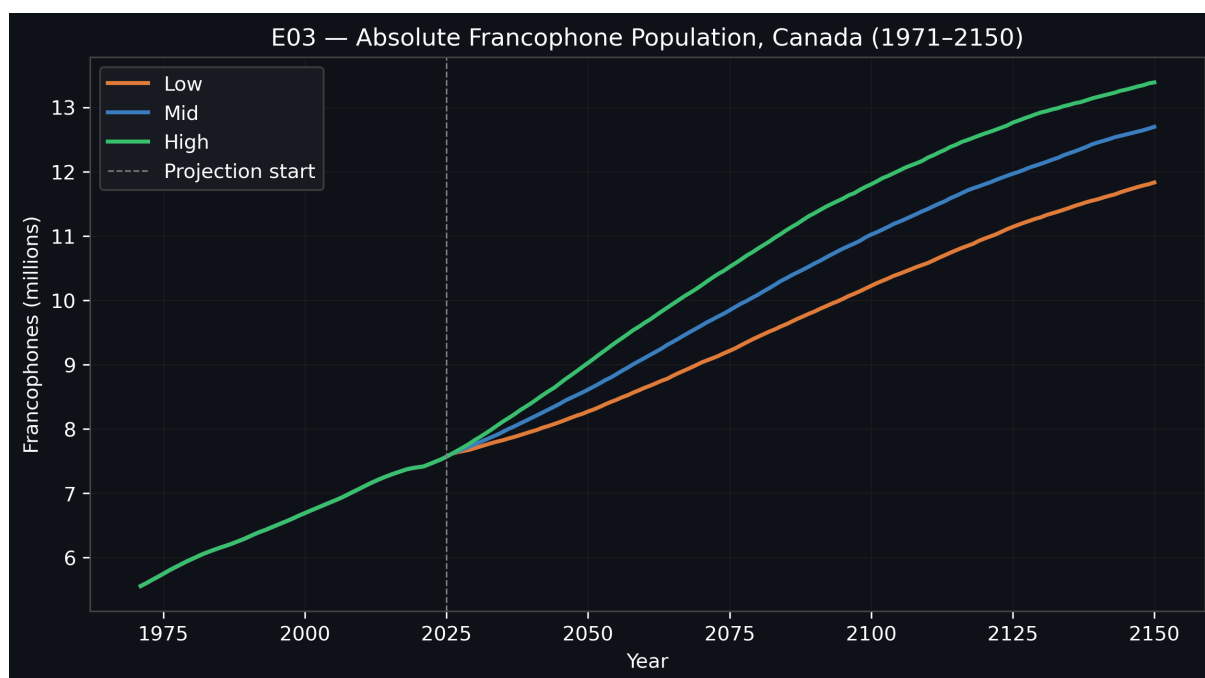


Figure 4. Total francophone population of Canada in millions (1971–2150). All three scenarios show continuous growth, reflecting the large base population and continued francophone immigration.

The francophone population grows from **7.5 million** in 2025 to **12–13 million** by 2150. The share decline is almost entirely a denominator effect: non-francophone Canadians grow considerably faster, not because there are fewer francophones, but because there are substantially more of everyone else.

This distinction is important for policy. Institutions serving francophone Canadians will face a *larger* absolute population even as that population represents a declining fraction of the total. The policy challenge is not protecting a shrinking community but maintaining relative influence and institutional capacity for a growing one.

4.4. Finding 4 — Quebec Holds; the Rest of Canada Does Not

Figures 5 and 6 show the divergence between Quebec and the Rest of Canada — the simulation's most consequential regional split.

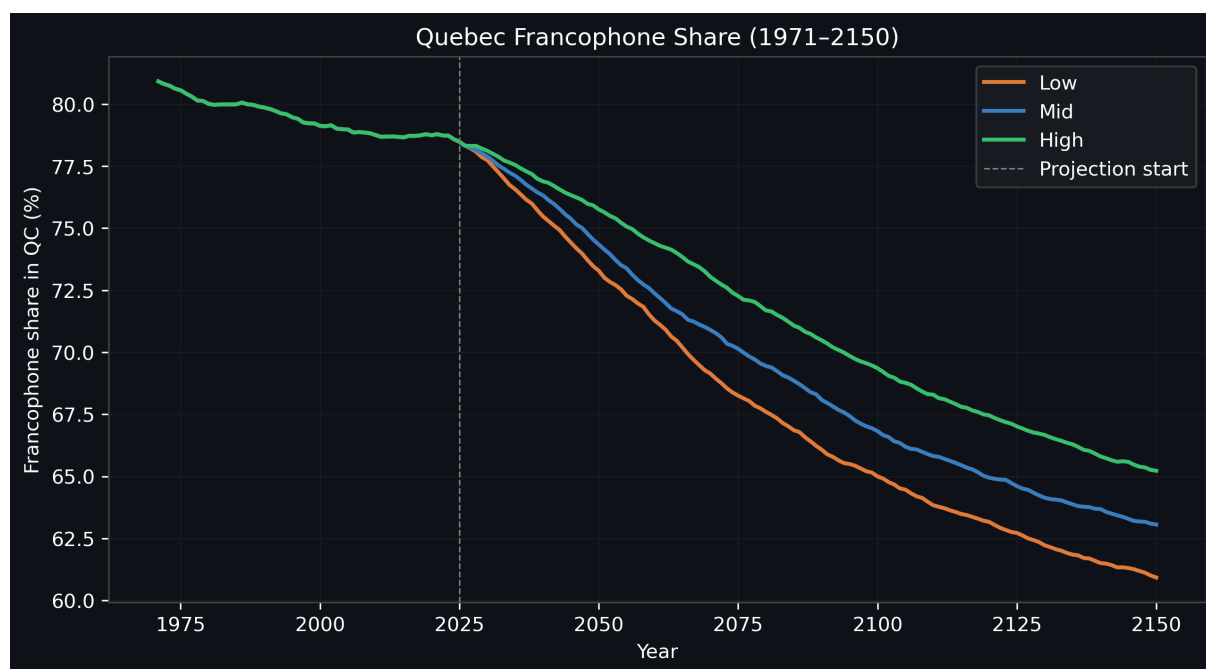


Figure 5. Quebec's internal francophone share (1971–2150). All scenarios project Quebec remaining above 70% francophone throughout, with a slow decline driven by growing allophone immigration.

Quebec's francophone share declines modestly but remains dominant across all scenarios. It falls from 77.5% in 2021 to approximately 65–72% by 2150, depending on scenario. Bill 101 continues to drive allophone integration into French, and Quebec's own selection process sustains a majority francophone immigrant intake.

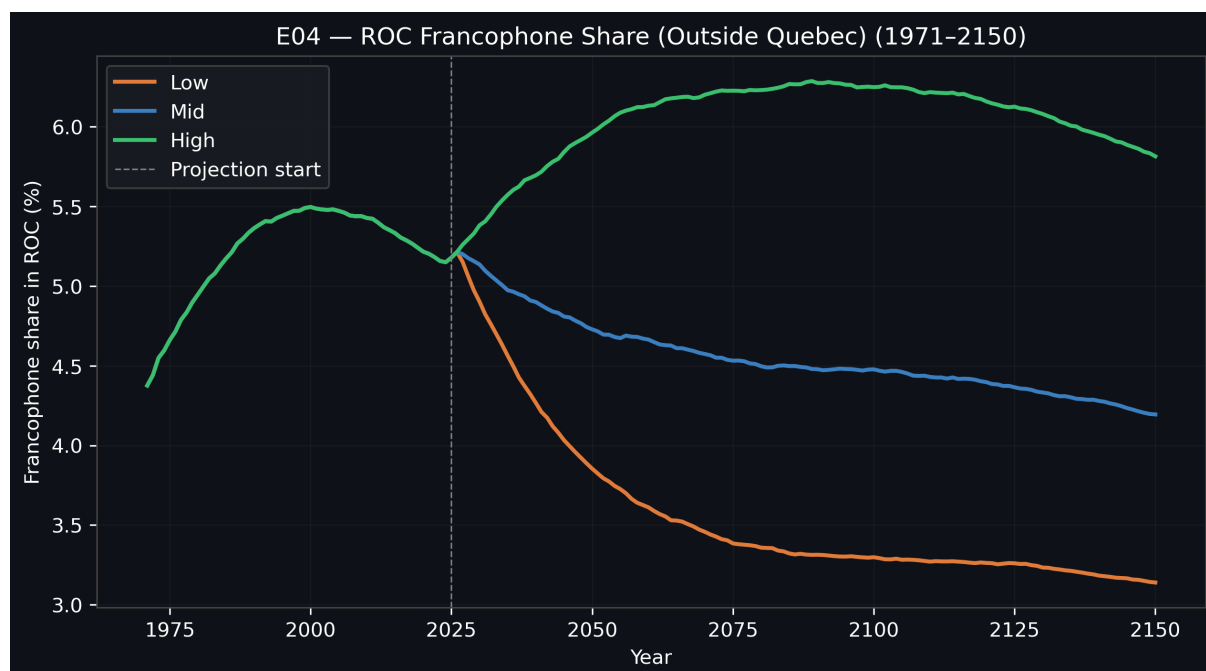


Figure 6. Francophone share of the Rest of Canada (outside Quebec), 1971–2150. The decline is considerably steeper than the national aggregate, driven by low francophone immigration shares and high French-to-English language transfer.

Outside Quebec, the picture is starkly different. The ROC francophone share falls from approximately 5.3% in 2025 to between 3.0% (Low) and 4.2% (Mid) by 2150. Only the High scenario — with its elevated ROC francophone immigration override — bucks the trend, rising briefly to 6.3% before declining.

The ROC decline is driven by two forces simultaneously: the same immigration-driven dilution present nationally, but with a far lower francophone intake fraction; and a language transfer rate from French to English ($\approx 1.8\%$ /year) that is almost 36 times higher than the equivalent rate in Quebec.

Key takeaway. Quebec's francophone majority is robust across all scenarios. The existential question for French in Canada is not Quebec — it is the survival of francophone communities outside it. ROC francophones are under pressure from both dilution and assimilation simultaneously.

4.5. Finding 5 — New Brunswick: Slow but Steady Erosion

New Brunswick is Canada's only officially bilingual province and home to the Acadian community, the largest francophone population outside Quebec. Figure 7 tracks NB language shares under all three scenarios.

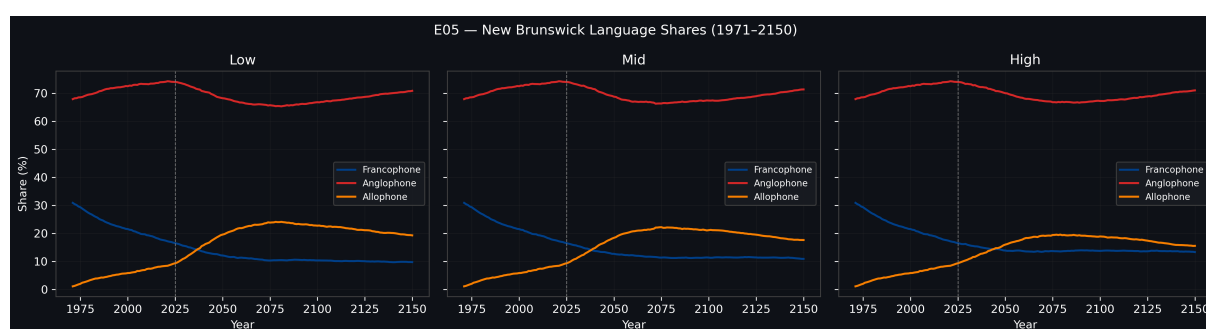


Figure 7. New Brunswick language shares under Low (left), Mid (centre), and High (right) scenarios. The francophone share (blue) declines steadily while the allophone share (orange) grows substantially from near zero.

The NB francophone share falls from 31% in 1971 to between 14% and 16% by 2150 — roughly halved. Anglophones remain the dominant group but also lose relative ground as allophones grow. The Acadian community's below-average out-migration rate (multiplier 0.70) and the province's bilingual institutions provide some protection, but the erosion is sustained across all scenarios.

4.6. Finding 6 — Immigration Dilution Drives the Decline; Language Transfer Is Secondary

Figure 8 decomposes the annual change in the national francophone share into two components for the Mid scenario: (1) the immigration dilution effect and (2) the combined effect of language transfer and natural increase.

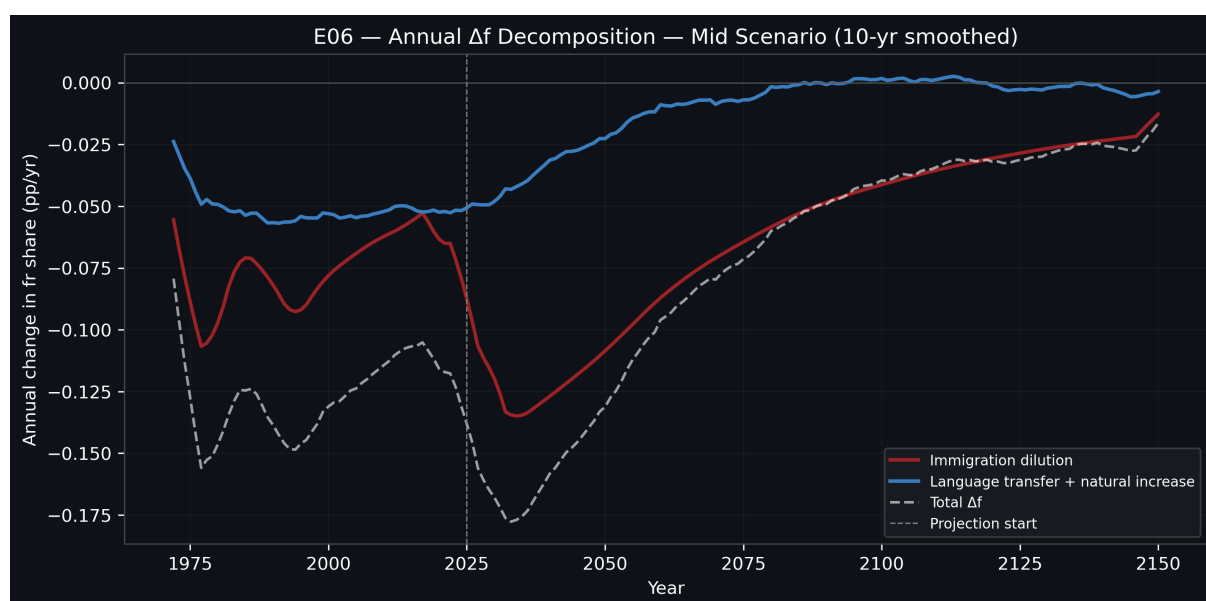


Figure 8. Annual change in national francophone share decomposed into immigration dilution (red) and language transfer plus natural increase (blue), Mid scenario, 10-year smoothed. Immigration consistently dominates the decline.

Immigration dilution is the dominant and persistent force throughout. The language transfer contribution is negative but smaller, and diminishes after 2050 as the natural increase term (which is language-neutral in the model) becomes negative across most provinces, partially offsetting assimilation losses. By 2100, the total annual change in share is less than 0.05 percentage points — with immigration accounting for nearly all of it.

Policy implication. Because immigration dilution dominates, policies that raise the francophone fraction of the immigrant intake have a far greater impact on the trajectory than programmes targeting language transfer or retention. A 5 percentage point increase in the national immigrant francophone share is worth more than eliminating French-to-English assimilation in the ROC.

4.7. Finding 7 — Stabilising the Share Would Require Doubling the Immigrant Francophone Fraction

Figure 9 plots the francophone share of immigrants that would be required each year to hold the national francophone share perfectly flat (solid lines), alongside the actual f_{imm} delivered in each scenario (dashed lines).

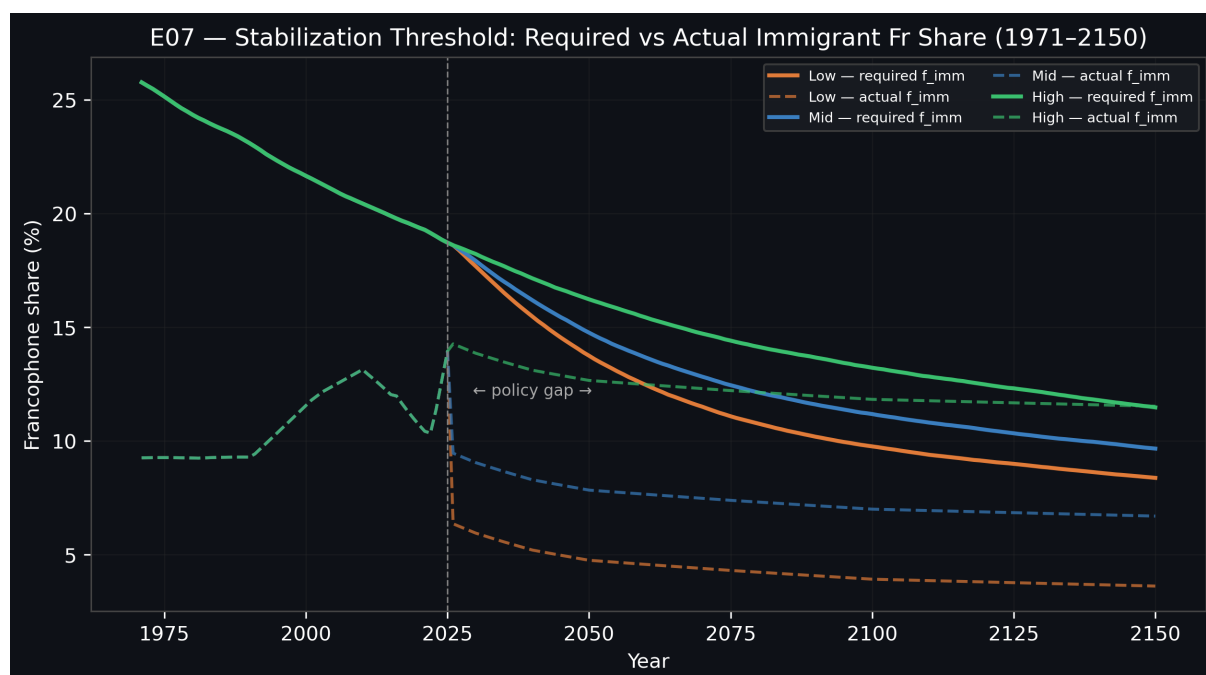


Figure 9. Stabilisation threshold (solid): the francophone share of immigrants required to hold $f(t)$ constant, equal to $f(t)$ itself each year. Actual immigrant francophone share by scenario (dashed). The gap between solid and dashed is the policy shortfall.

In 2025 the stabilisation threshold stands at approximately 19% — meaning 19% of all immigrants to Canada would need to be francophone to prevent any decline. The actual effective national f_{imm} is approximately 6% (Low), 10% (Mid), and 14% (High), leaving a policy gap of 5–13 percentage points.

As the share declines, the required threshold falls — by 2075 it is approximately 12% under the Mid scenario — and the High scenario comes closest to closing it. Even so, no scenario closes the gap within the projection window.

5. Conclusions & Strategic Implications

Summary of Conclusions

1. Francophone share decline is structural and cannot be reversed under any near-term realistic policy trajectory.
2. Composition of immigration matters more than volume.
3. The decline decelerates and does not produce demographic extinction.
4. Quebec's position is resilient; the ROC is the existential front.
5. The absolute francophone population grows, shifting the framing from survival to influence and institutional capacity.

Composition, not volume, is the primary policy lever. The simulation demonstrates clearly that increasing immigration volumes without raising the francophone share of arrivals worsens the outcome

for French. The Low scenario uses 10% higher volumes than the Mid scenario yet produces the worst results — because the composition disadvantage compounds the volume. Policy interventions aimed at increasing the francophone fraction of the intake (recruitment in francophone Africa, QC selection regime continuity, ROC francophone stream expansion) have a larger impact per dollar than equivalent investments in post-arrival language retention.

Quebec is not the concern; the ROC is. All scenarios project Quebec maintaining a francophone supermajority above 65% for the full projection period. Bill 101, Quebec's selection powers, and the extremely low out-migration of Quebec francophones together provide robust demographic protection. The crisis — if there is one — is in New Brunswick and the rest of francophone Canada outside Quebec. The ROC francophone share is projected to halve from $\approx 5\%$ to $\approx 3\text{--}4\%$ by 2150 even under the Mid scenario, driven simultaneously by dilution and English assimilation.

Absolute growth reframes the institutional challenge. The francophone population will grow from 7.5 million to 12–13 million by 2150. This is not a community in numerical decline. The challenge is one of relative political weight, institutional funding formulas, and cultural influence in a country that is growing faster still. Policies calibrated to a narrative of community survival may be less useful than policies calibrated to maintaining institutional capacity for a growing population whose share of national life is declining.

Time matters. The plateau mechanism means that the window to influence outcomes is concentrated in the near term. The gap between the current share and the immigrant fraction is widest now; each decade of underperformance on francophone immigration targets locks in a lower equilibrium. Conversely, sustained ambition in the near term has compounding benefits as the gap narrows and the system naturally stabilises at a higher floor.

Model limitations. The simulation does not model age structure, fertility differences by language group, socio-economic sorting, or policy feedback loops. All post-2025 inputs are projections, not forecasts. The model is a tool for understanding mechanisms and comparing scenarios, not for producing point predictions of future demographic outcomes.